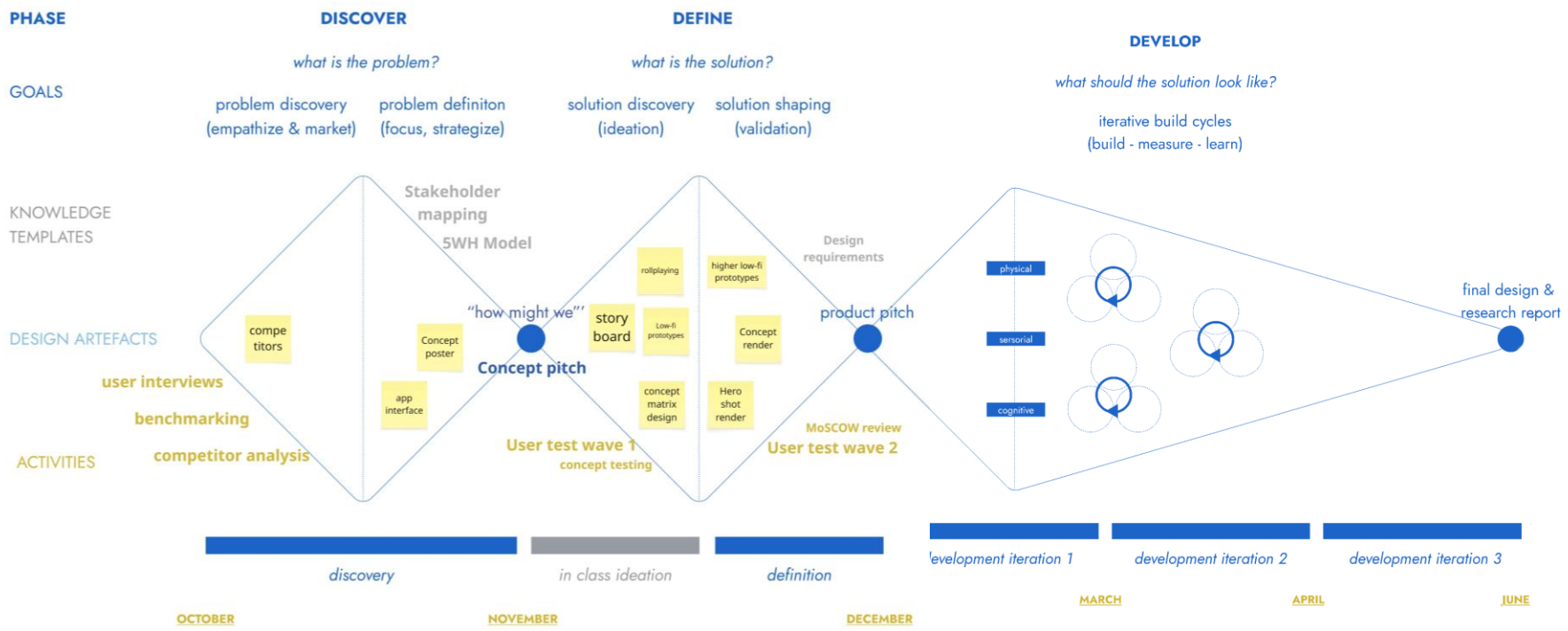
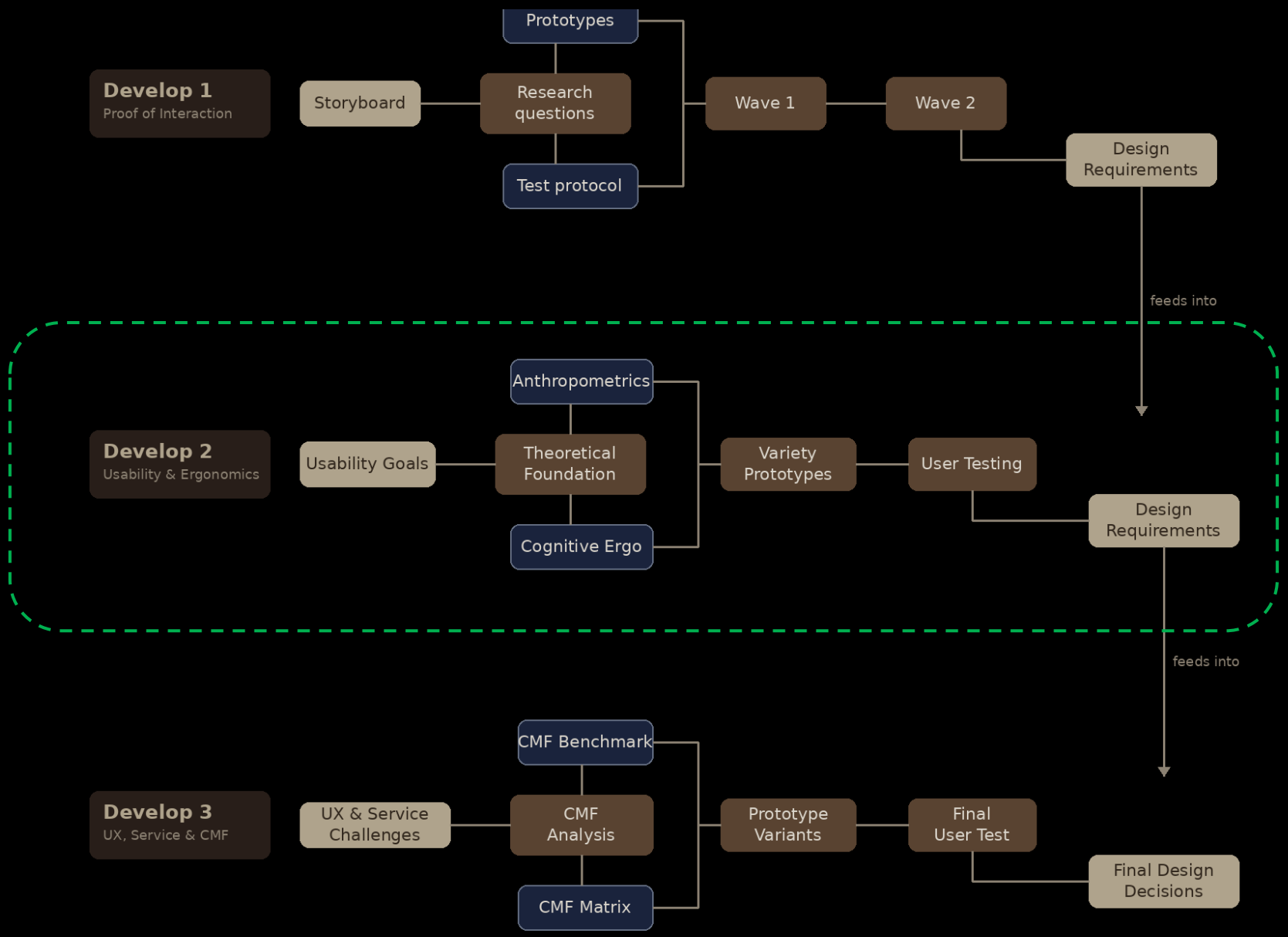
SensePath

Ear free, intuitive navigation in & outdoors


PHASE
DISCOVER
DEFINE
DEVELOP
GOALS
what is the problem?
what is the solution?
what should the solution look like?
problem discovery (empathize & market)
problem definiton (focus, strategize)
solution discovery (ideation)
solution shaping (validation)
iterative build cycles (build - measure - learn)
KNOWLEDGE TEMPLATES
Stakeholder mapping
5WH Model
Design requirements
DESIGN ARTEFACTS
compe titors
Concept poster
app interface
"how might we"
Concept pitch
story board
rollplaying
Low-fi prototypes
concept matrix design
higher low-fi prototypes
Concept render
Hero shot render
product pitch
physical
sersorial
cognitive
final design & research report
user interviews
benchmarking
competitor analysis
ACTIVITIES
User test wave 1
concept testing
MoSCOW review
User test wave 2
discovery
in class ideation
definition
development iteration 1
development iteration 2
development iteration 3
OCTOBER
NOVEMBER
DECEMBER
MARCH
APRIL
JUNE

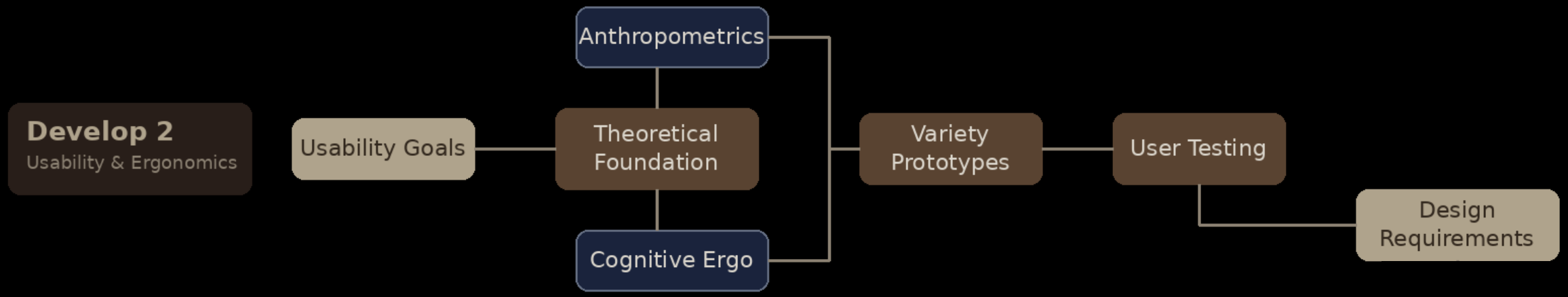
Develop

Methodology



Develop 2

Methodology

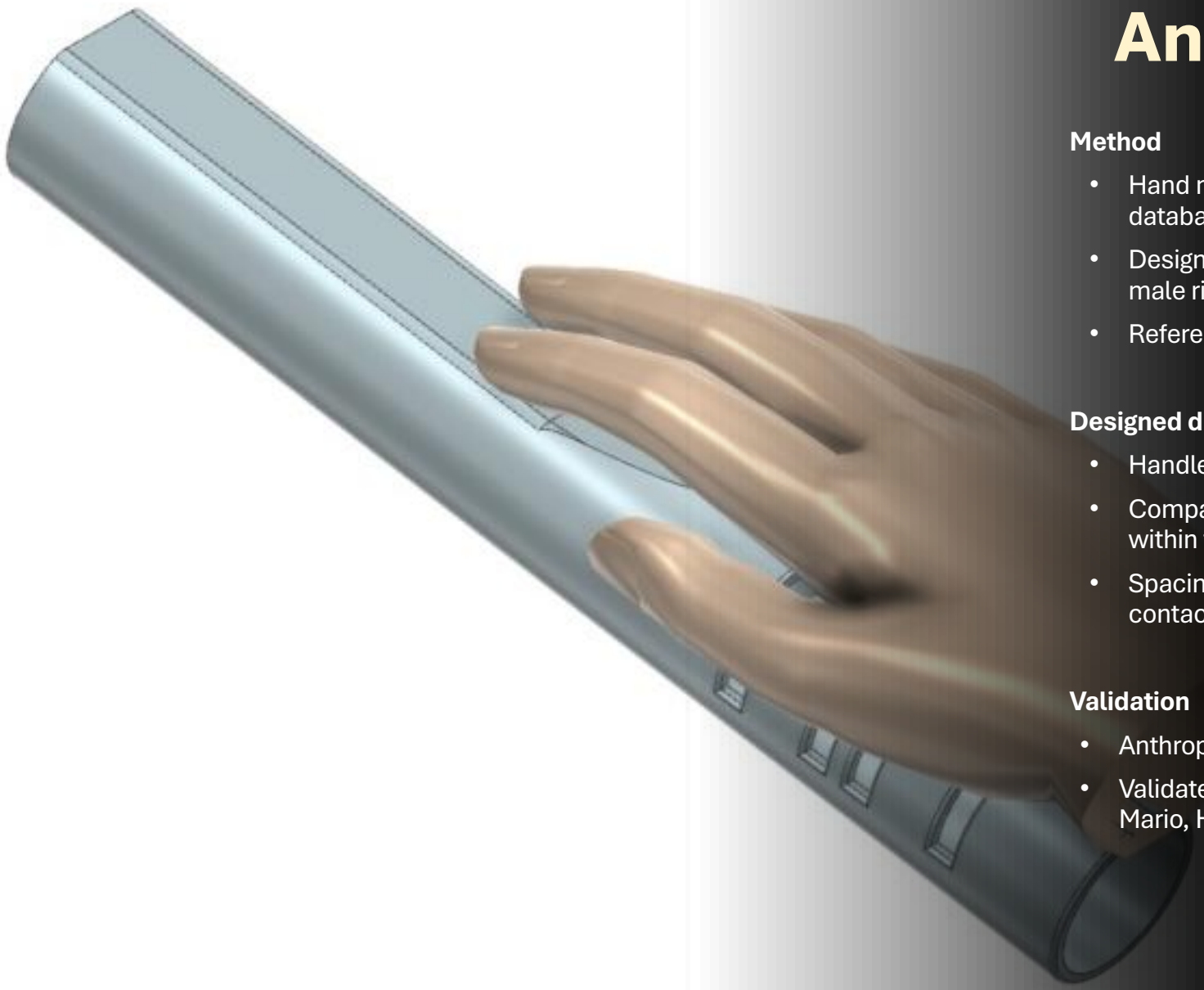


Usability goals

Effectiveness
UG1 - Path accuracy: SIZ >= develop 1 on the benchmark path *(observed)*
UG2 - Hesitation moments: <= number of hesitations in develop 1 *(observed)*

Satisfaction
UG3 - Cognitive load: Likert >= 4.5 *(perceived)*
UG4 - Trust : Likert >= 5 *(perceived)*
UG5 - Willingsness to use: Likert >= 5.5 *(perceived)*

Preference ranking
UG6 - Compass groove position: ranked preference across 5 positions *(perceived)*
UG7 - Compass element: ranked preference across 5 elements *(perceived)*



Anthropometrics

Method

- Hand model imported in **Siemens NX** via the **ANSUR II** database
- Design strategy: **middle 50% of the population** (P25–P75), male right hand, bare hands, finger-press posture
- Reference values: **length 19.3 cm / breadth 8.8 cm** (P50)

Designed dimensions

- Handle diameter calibrated to fit the middle 50% grip range
- Compass groove length sized so all 5 vertical positions fall within the active palm contact zone of the middle 50%
- Spacing between groove positions derived from finger-press contact mapping in NX

Validation

- Anthropometric output fed directly into the prototype CAD
- Validated empirically during user testing (Test 1 + Test 2) with Mario, Herman and Rory

Cognitive ergonomics

Starting point: what failed in Develop 1

- Combined condition: **97.52% accuracy but only 3.00/7 cognitive load**
- Two **haptic channels overlapped** on the same hand zone
- Compass mapping was **grip-dependent** (system image mismatch)
- Three **signifiers missing**: mode transition, system uncertainty

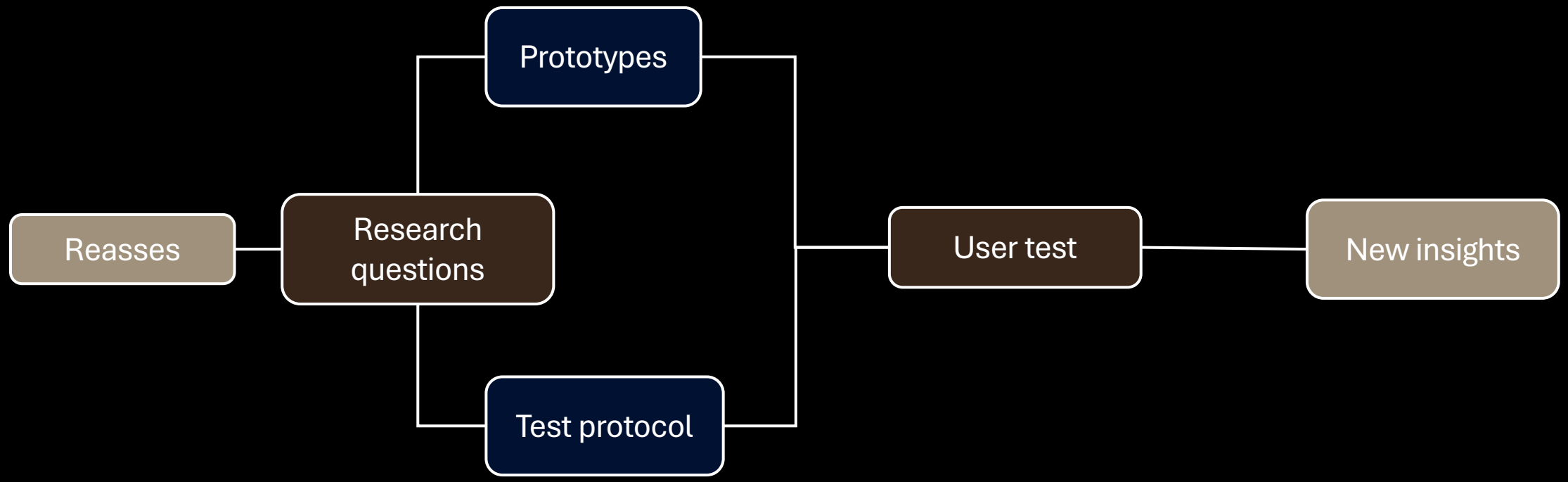
Theoretical lenses applied

- **Norman's 7 Stages of Action** → Perceive, Interpret and Compare were the broken stages
- **Gestalt laws** → figure-ground, continuity, proximity and common region applied to haptic perception
- **Signifiers** → in a non-visual interface, every status change must be tactile

Translation into the prototype

- **Spatial separation of channels** → finger-mounted vibration motors removed, single in-handle motor with a limited code set
- **Mental mapping safeguarded** → onboarding tutorial calibrates orientation to the user's natural grip; IMU fallback alerts the user when grip rotation breaks the mapping
- **Figure-ground contrast** → vibration is short and powerful; compass pauses during a vibration cue
- **Two new signifiers** → directional pre-announcement, outside→inside transition, IMU grip-rotation fallback

Methodology



Research Questions

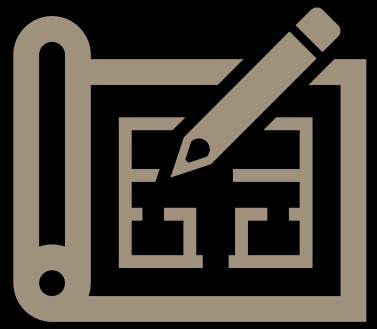
- At what height within the handle groove is the compass perceived most clearly?
- Which contact surface of the compass (rounded rectangular, spherical element, or angled rounded transitional shape) conveys directional movement most clearly?
- Does the preferred configuration reduce perceived cognitive load and increase confidence compared to Develop 1?
- Does path accuracy (SIZ) remain at least at the Develop 1 level with the new compass position and adjusted feedback?

Goal

Optimize **ergonomics**

Reduce **cognitive load**

Benchmark develop 1

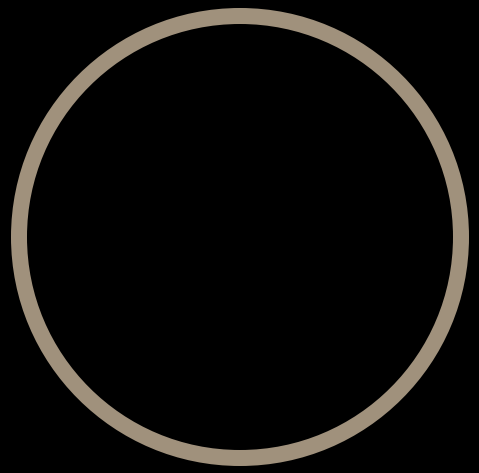


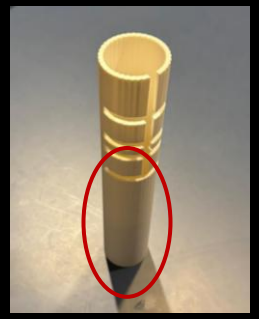
Define **design requirements**

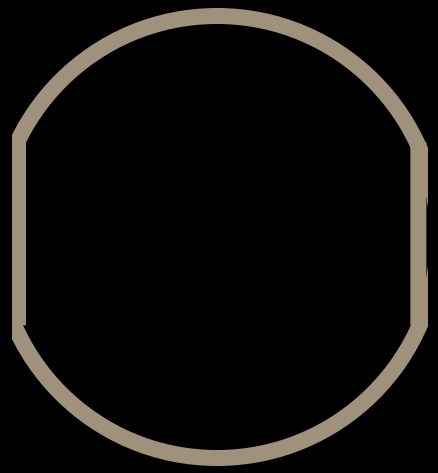
Sample N=5

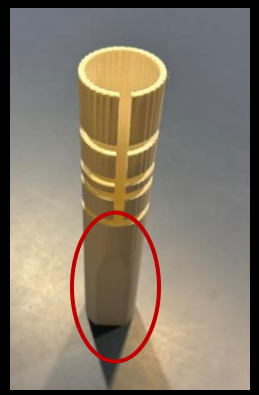
Name	Age	Relevance	Location	Time
Mario	52	Lost vision at age 18	Licht en liefde	07/04/2026
Rory	55	Severely visually impaired	Licht en liefde	09/04/2026
Herman	65	Blind since birth	Licht en liefde	08/04/2026
Milo	20	Sighted (control)	At home	10/04/2026
Milos	20	Sighted (control)	At home	10/04/2026

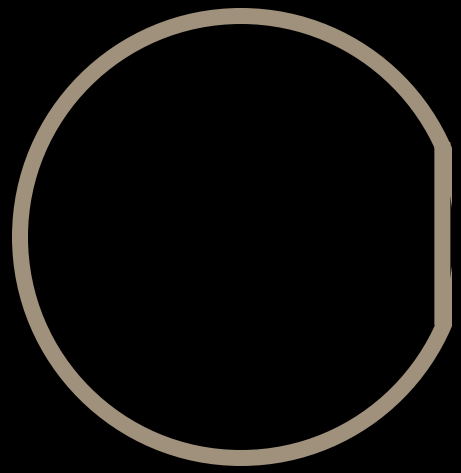
Prototype: Handle profile

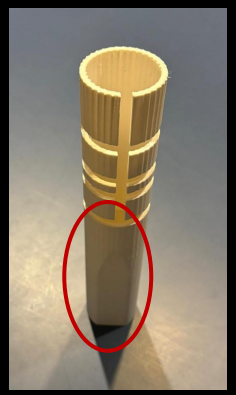






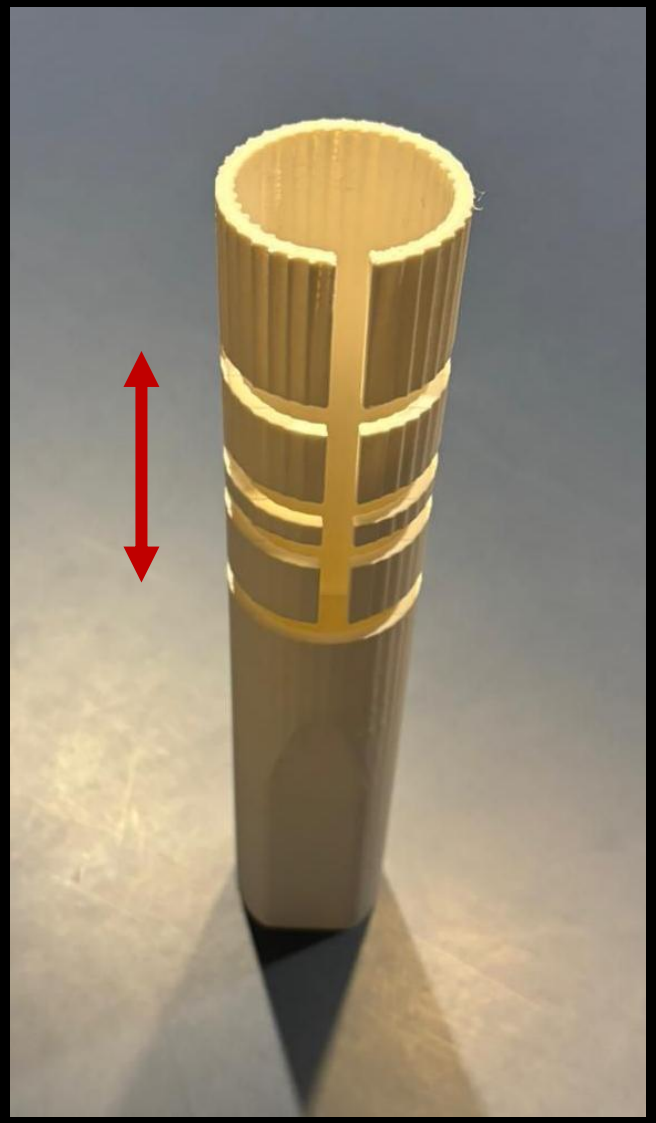






Prototype: Compass groove

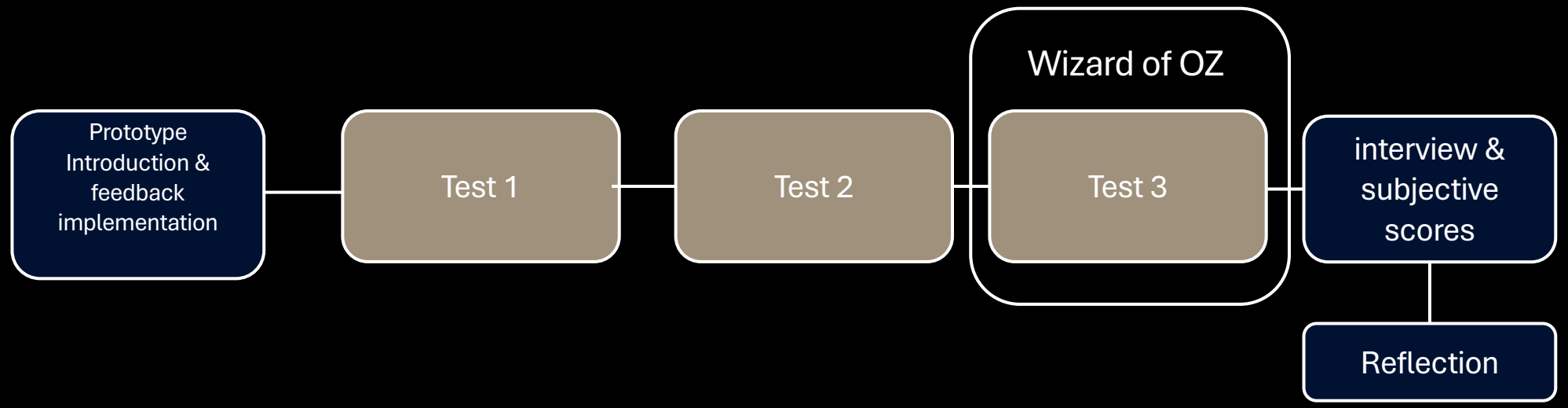
- Prototype that tests different possible grove positions to see which one feels best



Prototype: Contact surface

- Prototype that tests different contact surfaces to see which one feels best

Workflow


Prototype Introduction & feedback implementation
Test 1
Test 2
Wizard of OZ
Test 3
interview & subjective scores
Reflection

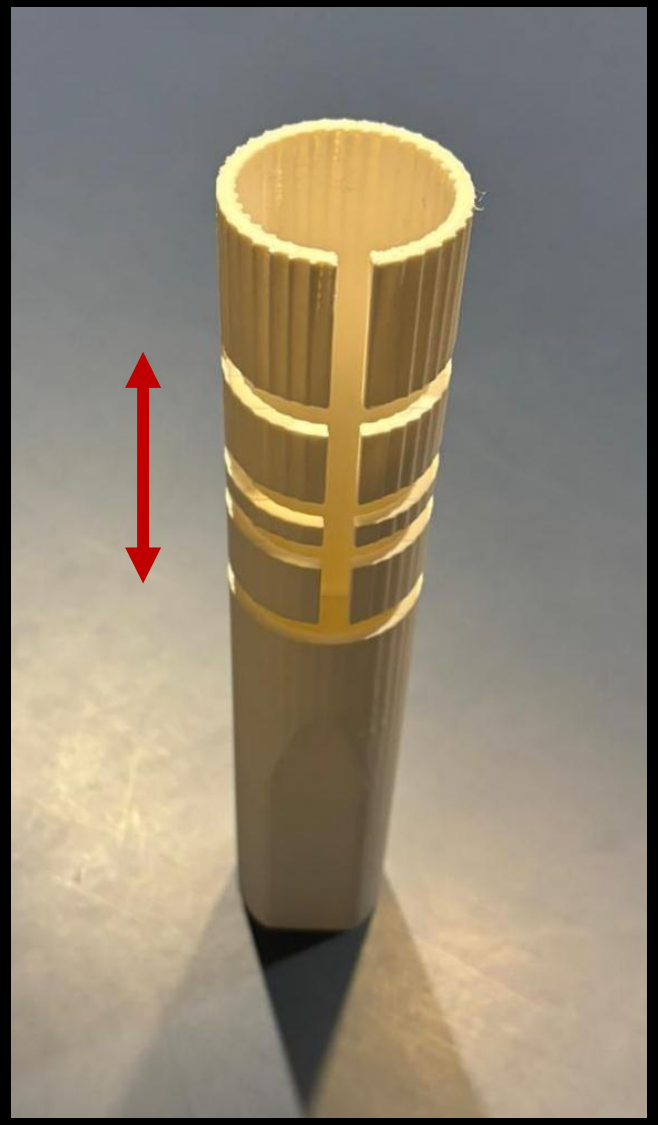
Test 1: Compass grooves

What we tested

- **Goal** → find the groove position on the handle where users perceive the compass element most clearly, given their own natural grip
- **Variants** → 5 vertical groove positions along the handle (H1 lowest, near the wrist, to H5 highest, near the knuckles)
- **Method** → participant feels each position in standstill, then walks a short straight strip with the compass element placed in that groove
- **Reference surface** → the rounded rectangular protrusion at medium height was kept constant across all 5 positions, so only groove position varied

What we found

- **Lowest groove (H1) preferred** → both Mario and Herman independently picked the position closest to the heel of the hand
- **Hypothenar contact zone** → at the lowest position the element rests against the firm fleshy part near the wrist, where the hand naturally maintains contact regardless of grip
- **Higher positions problematic** → the topmost position caused skin to get pinched between the element and the groove edge
- **Grip-independent** → the lowest groove position works across different grip styles because the contact surface is anatomically anchored



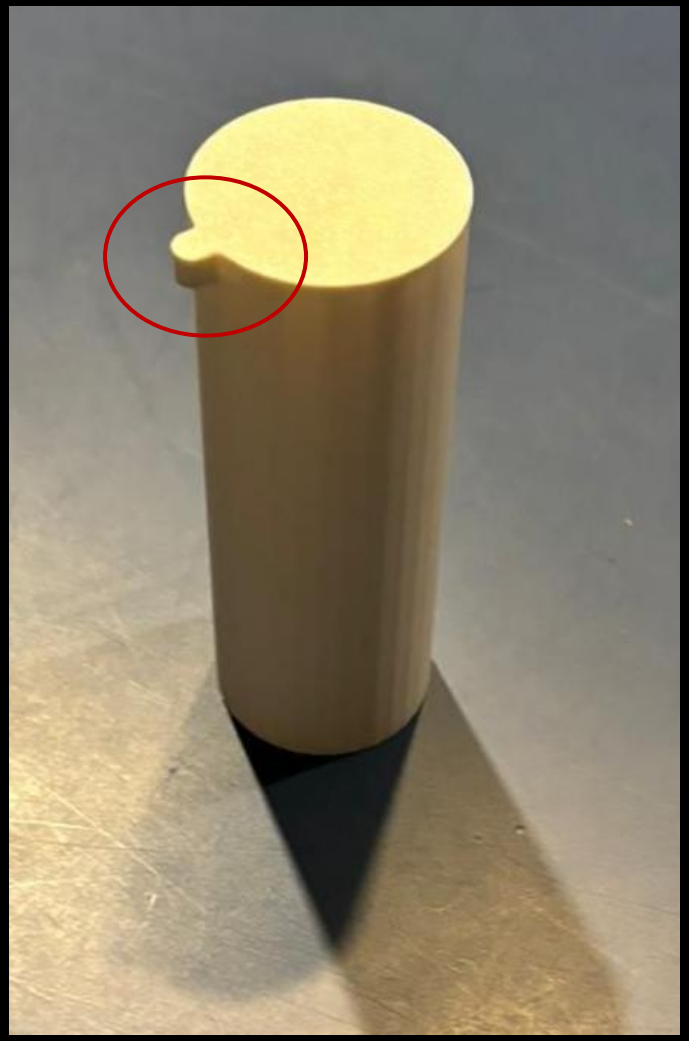
Test 2: Contact surface

What we tested

- **Goal** → find the compass element shape and protrusion height that conveys directional movement most clearly and comfortably, on the preferred groove position from Test 1
- **Variants (5 elements)**
  - C1: Rounded rectangular protrusion, **low** height
  - C2: Rounded rectangular protrusion, **medium** height
  - C3: Rounded rectangular protrusion, **high** height
  - C4: **Spherical** element
  - C5: **Angled rounded** transitional shape (gradual ramp from flat to raised, added after Mario's insight in session 1)
- **Method** → participant feels each element in standstill, then walks a short straight strip per variant; final ranking by participant
- **Two dimensions varied together** → both shape and protrusion height, since both influence perceived clarity and comfort

What we found

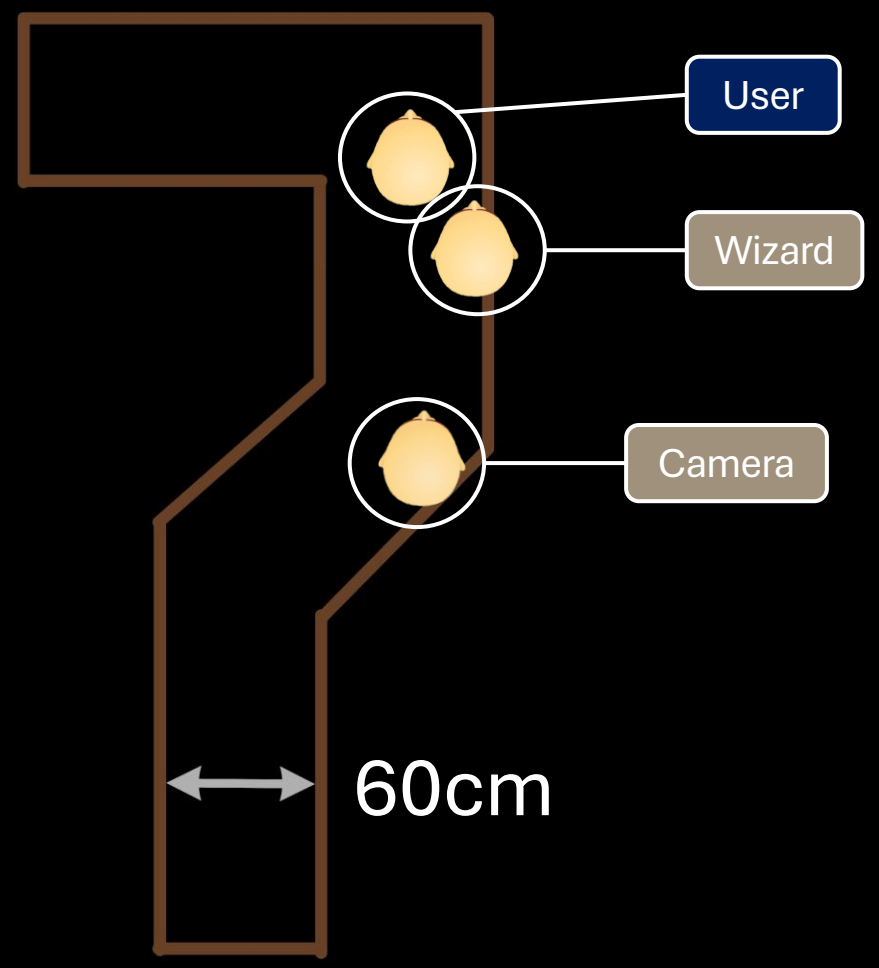
- **Medium protrusion height won** → low felt too subtle, high felt sharp and uncomfortable; medium gave the best balance between perceptibility and comfort
- **Spherical element preferred shape** → smoother directional transmission, no sharp edges against the skin, comfortable across different grip rotations
- **Angled rounded shape (C5) close second** → especially appreciated for the gradual transition from flat to raised, which Rory confirmed
- **Glove insight from Mario** → triggered the idea of an interchangeable compass module with a more robust winter variant



Path test 3

During the test

Test 3: Turn via compass feedback



- Wizard provides continuous feedback via handle compass

Test 3: Feedback

- **Single focus point is a real relief** → users only have to attend to the compass, no longer restricting hand position to feel finger-mounted vibration motors; the grip can stay natural
- **180° deflection is unnecessary in practice** → Mario noticed mid-walk that the system never needs that much rotation because it corrects continuously, even before he reached the corner
- **Hand-attention bottleneck only shows under realistic conditions** → Herman acknowledged the prototype works in this setting, but warned that with a real stick tapping the ground, kerb-following and ambient listening, the constant hand focus would compete with everything else

“It works, but my main concern stays the same: you constantly have to focus on your hand. For me personally, audio still works better”

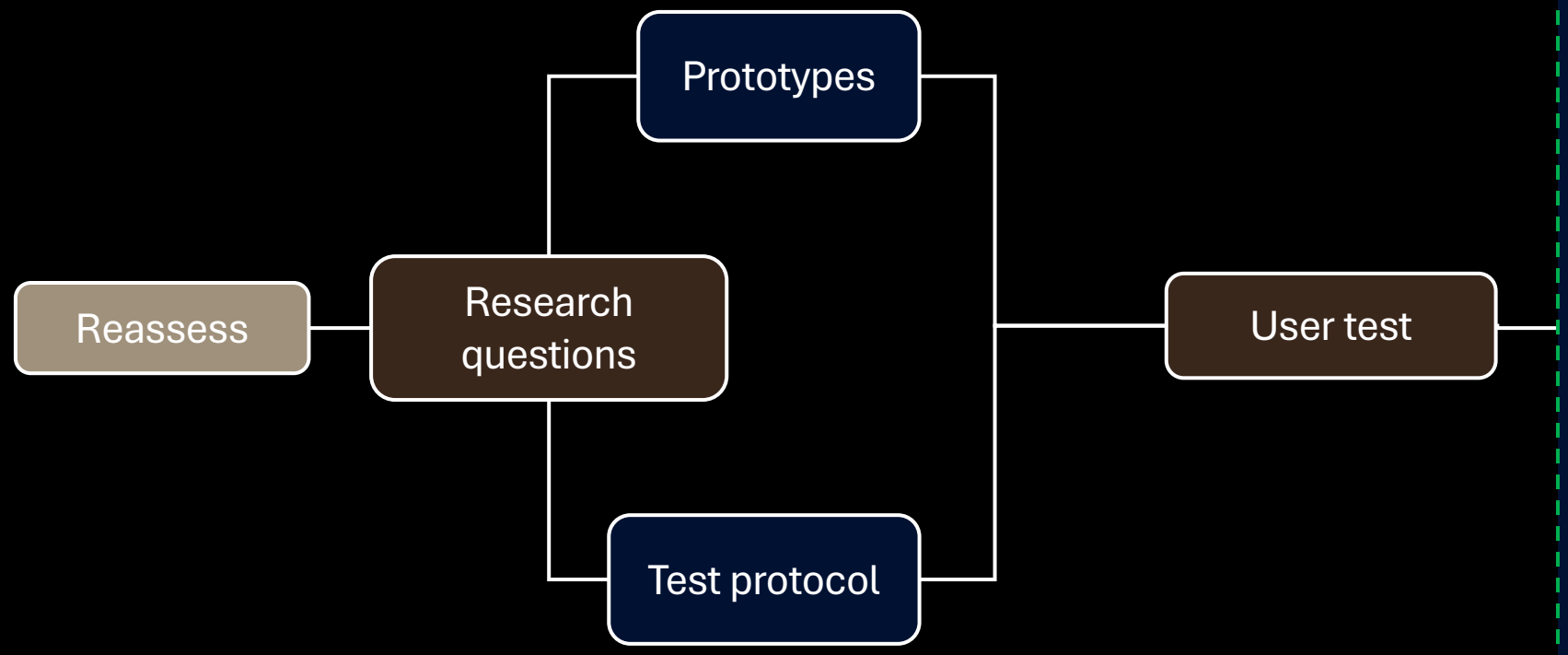
“I'm actually impressed, this test feels noticeably better than the previous one”

“180 degrees of rotation isn't really needed, the system corrects continuously, so a smaller range is more than enough.”

Test 3: Benchmark comparison

Metric	Develop 2	Develop 1
Walking zone width	60 cm	60cm
TIZ (% Time in zone) VS SIZ (% Steps in zone)	100%	97,52%
Apprenticeship	Av. 4 min	Av. 4min
Learnability (1–7)	6,33	6,00
Smoothness (1–7)	5,67	4,67
Cognitive load (1–7)	4,67	3
Willingness (1–7)	4,33	4,33
Trust (1–7)	n.v.t.	n.v.t
Positioning system	Video+Observation	Video+Observation
Stick used?	No (loose handle)	Yes
Context	Interior space licht en liefde	Interior space licht en liefde

Develop 2 conclusions



- A **single in-handle vibration motor** with limited code set removes the grip-dependency.
- The **lowest groove position** is the optimal location for the compass element
- A **spherical contact surface at medium height** delivers the clearest directional perception.
- **An interchangeable compass module** is next, enabling personal preference and a robust winter variant for glove use.
- **Cognitive load drops by 1.67 points** on a 7-point scale, the strongest evidence that ergonomics directly reduces mental effort.

Design requirements

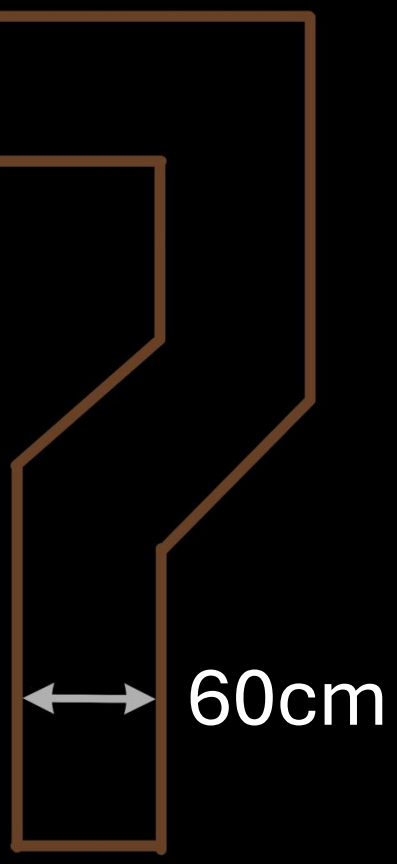
ID	Requirement
D6.1	The compass element shall be positioned at the lowest groove position on the handle (close to the hypothenar) for optimal tactile perception
D6.2	The compass element shall use a spherical contact surface as the default geometry
D6.3	The compass element shall have a medium protrusion height: high enough to be clearly felt, low enough to avoid skin pinching or sharp pressure
D6.4	The compass shall behave as a true compass: full deflection toward the target direction, gradual return to neutral as the user reorients
D6.5	The compass shall pause briefly during a vibration cue to preserve figure-ground separation between the two haptic channels
D6.6	The vibration set shall be limited to a small number of short, powerful codes (directional pre-announcement, outside to inside transition, fallback)
D6.7	The compass module shall be interchangeable, allowing users to swap to a more robust variant for glove use in cold conditions
D6.8	The handle dimensions shall accommodate the middle 50% of the target population (ANSUR II, P25 to P75) as validated in NX
D6.9	The system shall provide an IMU-based fallback signal when the user's grip rotation breaks the compass mapping, communicated as a distinct vibration pattern
D6.10	No vibration motors shall be placed on individual finger positions, to remove dependency on grip pattern variability

Critical reflection

Path change

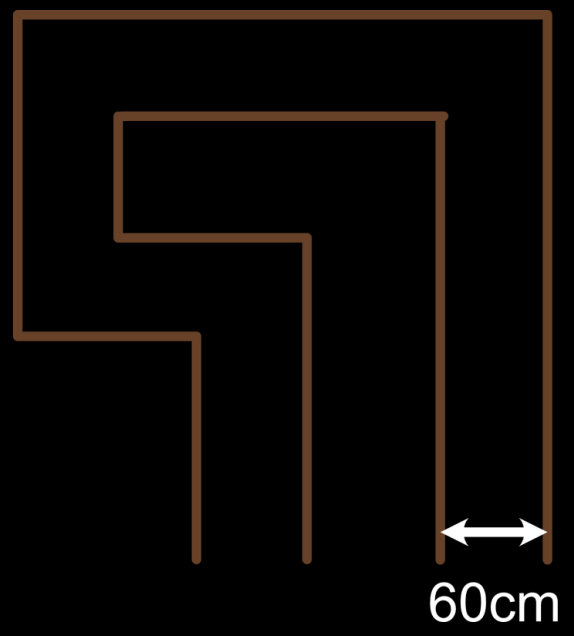
Test dev 1

Test dev 2

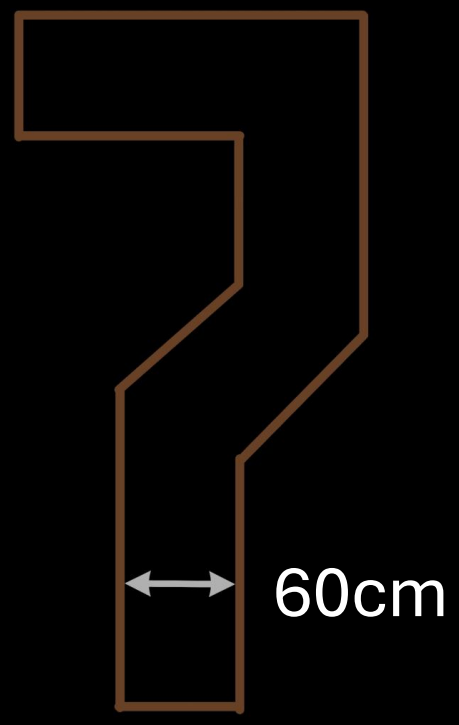


Shorter path

Path Learning



Develop 1



Develop 2

Users will unconsciously learn to navigate the path

References

- **Liu, G., Yu, T., Yu, C., Xu, H., Xu, S., Yang, C., Wang, F., Mi, H., & Shi, Y. (2021).** Tactile Compass: Enabling visually impaired people to follow a path with continuous directional feedback. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. ACM. https://doi.org/10.1145/3411764.3445644
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